

DataSentinel Data Origins & Lineage Report

1. Real Collected Source Data (Immutable)

The DataSentinel system heavily relies on true operational data for its base telemetry and observability logic. The following data is REAL and collected directly from source:

- Master Data / Claims: The D:\UC10_Data_Quality_Pipeline\master_data\ directory serves as the definitive repository of raw healthcare transactions (claims, authorizations, billing) from hospital providers.
- Pipeline Telemetry: As the pipelines process the raw master data, they emit REAL execution telemetry (throughput, records in/out, latency) stored in DuckDB and SQLite.
- Operational ML Outcomes: The Isolation Forest and CUSUM models evaluate true operational statistics, generating authentic anomaly scores based strictly on real processed data.
- Alert Engine: Most alerts surface naturally from real SLA breaches or data quality faults.

2. Synthetic & Injected Test Data

To ensure the UI and backend logic properly handle edge cases and edge incidents, the following data is deliberately synthetic and injected into the pipeline:

- Adversarial Data Files: Test suites inject synthetic malformed records (e.g. test_adversarial_records.csv) to trigger Data Quality and Anomaly checks explicitly.
- Manual Test Alerts: The alert-engine accepts direct JSON payloads via scripts like 'send_test_alert.py'. These are synthetic operational incidents (e.g. 'Claims volume is 60% below normal') strictly for UI deduplication and integration testing.
- Simulated Latency: Certain test cases synthetically stall pipeline events to artificially trigger the SLA Calculator for AT_RISK or BREACHED states.

3. Observability Guarantee

During Phase 7, the DataSentinel Frontend is strictly forbidden from fabricating its own data. The UI acts as a pure lens over the Backend Database.

If a metric is 'UNKNOWN' or 'NOT AVAILABLE' in the real telemetry, the UI will report it as such, rather than masking the void with synthetic information.